

# KNN Classification - Interview Questions & Answers

***Q: Why is KNN called a Lazy Learner?***

A: KNN does not build a model during training. It stores training data and makes predictions only when a new sample arrives.

***Q: How do you choose the optimal K value?***

A: Use the Elbow Method or cross-validation to find the K that gives the best performance.

***Q: What is the impact of feature scaling on KNN?***

A: KNN relies on distance calculations, so scaling ensures all features contribute equally.

***Q: What are common distance metrics used in KNN?***

A: Euclidean, Manhattan, Minkowski, and Cosine distance.

***Q: What are the advantages of KNN?***

A: Simple, easy to understand, no training phase, and works well for small datasets.

***Q: What are the disadvantages of KNN?***

A: Slow on large datasets, sensitive to noise, and requires feature scaling.

***Q: Difference between KNN Classification and Regression?***

A: Classification predicts categories; Regression predicts continuous values.

***Q: What is the Elbow Method?***

A: A technique to choose the best K value by analyzing error rates for different K values.

***Q: What is overfitting in KNN?***

A: Very small K values can memorize training data and perform poorly on new data.

***Q: How does KNN make predictions?***

A: It finds the K nearest neighbors and predicts based on majority voting.

***Q: What is the role of Euclidean Distance?***

A: It measures similarity between data points and helps identify nearest neighbors.

***Q: Why is train-test split important?***

A: It evaluates model performance on unseen data.

***Q: What is the ROC Curve?***

A: A graph of True Positive Rate versus False Positive Rate.

***Q: What does AUC represent?***

A: Area Under the Curve; higher values indicate better classification performance.

***Q: When should KNN not be used?***

A: For very large datasets or high-dimensional data where prediction becomes slow.